

What Are We Measuring? Benchmark Integrity and a Zone-Based Reframing for Emotion Recognition in Autistic Children

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Abstract—Automated emotion recognition is widely proposed as an assistive technology for autistic children, and recent papers report 97.95–99.99% accuracy on public autism facial-expression benchmarks. Independently collected clinical corpora report 40–78% on the same task. We resolve this contradiction by auditing the most widely used public archive. Re-acquiring it and verifying it byte-for-byte against our earlier run, we find that it ships two overlapping dataset trees sharing 45% of the smaller tree’s images, that 21.4% of its files are byte-identical duplicates of another file, and that 119 duplicate groups carry mutually contradictory emotion labels on pixel-identical images. Holding features, classifier, and label space fixed and varying only the split protocol, six-category balanced accuracy spans 0.325 to 0.709 across five defensible readings of the same archive. Duplicate leakage itself proves second-order, worth 4.4 points, and no protocol we construct approaches the published figures; the dominant effect is which subset one evaluates on. The label contradictions are not uniform: 279 of 280 zone-level conflicts fall in negative affect and one falls in positive/neutral. Annotators agree on valence and cannot agree within negative valence, so the six-category schema is itself manufacturing the noise it is scored against. Collapsing to three action-oriented Emotion Zones cuts protocol-induced variance by 38%, which argues for coarser labels and multimodal context over finer face-only categories. We report chance-corrected metrics throughout so a collapse to fewer classes is never credited for the chance floor moving. This is a measurement audit, not a deployable classifier, and not a claim about any child’s internal state.

Index Terms—benchmark integrity, data leakage, label noise, affective computing, autism

I. INTRODUCTION

Roughly 1 in 31 eight-year-olds in the United States is diagnosed with autism spectrum disorder (ASD) [1]. Because emotional expression in autistic people is frequently misread by non-autistic observers—the *double empathy problem* [2]—

automated emotion recognition is repeatedly proposed as a communication aid for autistic children and their caregivers.

The literature on this task is in an unusual state. Papers evaluating on public autism facial-expression archives report accuracies of 97.95% [3], 99.8%, and 99.99%. Papers evaluating on independently collected, consented clinical corpora report far less: 40% for a commercial analyser on the EMBOA autism corpus [4], 57.95–64.92% for four commercial classifiers on parent-reported autistic children [5], and 66.43% for a classifier trained on typically developing children and tested on autistic children [6]. The gap between these two literatures is roughly 30 accuracy points, and it tracks how the evaluation data were obtained rather than what the models are.

That correlation has recently become impossible to ignore. Between November 2025 and 2026, publishers retracted or flagged roughly 90 papers built on scraped autism face datasets, including the source paper for the archive we audit here [7] and a facial emotion recognition paper [8], on the grounds that the images were collected from the internet without documented diagnosis, ethical oversight, or guardian consent. That established the *provenance* failure. It did not establish whether the reported numbers were also, independently, measurement artifacts.

This paper supplies that missing measurement. Our contributions are:

- 1) A quantified integrity audit of the most widely used public autism emotion archive, computed from the vendor’s own directory layout, reporting duplicate structure, cross-copy overlap, and cross-label duplicate groups (Sec. III).
- 2) A leakage ablation that holds features, classifier, and label space fixed and varies only the split protocol across five plausible readings of the archive, separating protocol effects from signal (Sec. IV).
- 3) The observation that the label contradictions are concentrated almost entirely in negative affect, which we argue is evidence that the categorical schema is generating them (Sec. V).
- 4) A resulting argument for coarser action-oriented zones and multimodal context over finer face-only categories,

with chance-corrected metrics so the collapse is not credited for the chance floor (Sec. VI).

We are auditing an instrument, not building one. We make no claim about any child’s internal state, and we treat every label as an annotator judgment about an image rather than as ground truth about a person.

II. RELATED WORK

Cross-population transfer is established. Grossard *et al.* [6] ran the controlled experiment on lab-collected video from 157 typically developing and 36 autistic children with subject-independent cross-validation, obtaining 82.05% within-population and 66.43% when training on typically developing children and testing on autistic children. Gaya-Morey *et al.* [9] repeated the design with twelve CNNs on an adjacent population, observing roughly 90% within-population against under 55% across. We do not claim this finding. We ask a different question: whether the public benchmarks now used to measure it can support the measurement at all.

Benchmark contamination is a mature field. Barz and Denzler [10] found 3.3% and 10% duplicate rates in CIFAR-10 and CIFAR-100 test sets and released cleaned splits. Northcutt *et al.* [11] showed that label errors in test sets reorder model rankings. Ramos *et al.* [12] give the taxonomy we adopt: *hard* leakage for identical images across a split, *soft* for near duplicates, *intra-dataset* for overlap within one dataset’s own splits. Adimoolam *et al.* [13] name *augmented duplicates* and report 89% duplication in a geospatial benchmark. Apicella *et al.* [14] formalise *synthesis leakage*, where train and test items derive from a common source under a generating transform. None of this literature has examined a clinical affect dataset.

Contamination in this specific archive has been asserted, not measured. The authors of the FERAC derivative [15] reported “class overlap and image duplication” in this archive, removed duplicates, and dropped two classes as unrecoverable. They did not report a detection method or threshold, a duplicate rate, a cross-split leakage rate, or the effect of cleaning. Two independent groups have now flagged the same archive; what remains open is the quantity.

Why the labels might be the problem. Barrett *et al.* [16] challenge the premise that emotional state is readable from facial configuration at all. Keating *et al.* [17], using over 265 million motion-capture points across 4,896 expressions, find autistic and non-autistic expressions differ in which muscles carry each emotion and in kinematic smoothness, concluding the two groups may be “essentially speaking a different language.” Zane *et al.* [18] report that autistic expressions are produced at intensities non-autistic raters find ambiguous. If the expressions are ambiguous under a non-autistic reading, forcing them into six mutually exclusive categories should produce exactly the contradictions we measure.

Ethical framing. Nagy [19] argues that autism has been mobilised by emotion-AI developers as a charismatic use case and a testbed. We take that critique as a constraint on what this paper may claim.

TABLE I

THE TWO DATASET TREES SHIPPED IN ONE ARCHIVE.

Tree	Files	Unique	Train	Test
Balanced copy	1,425	1,380	1,199	226
Imbalanced copy	833	717	758	75
Shared unique images: 323 (45.0% of smaller tree)				

III. THE ARCHIVE

We audit the public Kaggle archive `fatmamtalaat/autistic-children-emotions-dr-fatma-m-talaat`, the dataset associated with [7]. We re-acquired it and verified that all 1,672 SHA-256 digests recorded in our earlier manifests are present in the fresh download, so the audit and the model results in Sec. IV describe the same bytes. All statistics below are computed from the vendor’s own directory layout, before any re-splitting: this is what a downstream user receives.

The archive ships two overlapping datasets. It contains two top-level trees (Table I) with different class balances and different train/test splits. They share 323 unique images, 45.0% of the smaller tree. This is a sufficient explanation for a puzzling feature of the literature: published descriptions of “the Talaat dataset” report its size variously as 348, 830, 1,603, and “2500+” images. Papers are describing whichever tree they loaded.

One file in five is a duplicate. Of 2,258 labelled images, only 1,774 are distinct by SHA-256; 484 files (21.4%) are byte-identical copies of another file. Duplicate groups reach size 5. Perceptual hashing at Hamming distance 4 adds 163 further near-duplicate components. A filename-suffix pattern (`_aug_N`) identifies 1,020 augmentation families, 335 of them larger than one and the largest containing 9 members, so a substantial part of the archive is synthetic variation on a smaller set of source images. This is synthesis leakage in the sense of [14], and unusually it is *announced in the filenames*: it was trivially detectable and shipped regardless.

Pixel-identical images carry contradictory labels. Under the vendor’s six categories, 119 byte-identical duplicate groups (327 files) contain members filed under more than one emotion. This is a stronger and cheaper claim than label-error detection in the sense of [11]: establishing it requires no external ground truth and no annotator panel, only a hash. It is an internal inconsistency of the artifact.

IV. HOW MUCH OF THE NUMBER IS THE PROTOCOL?

Contamination matters only if it moves results. We therefore hold the features, the classifier, and the label space fixed and vary *only* how train and test are separated. Features are frozen ViT-B/16 embeddings [20]; the head is a one-hidden-layer MLP with epoch-wise early stopping; every protocol is run over ten seeds [21]. The five protocols are all defensible readings of the archive: the two trees’ own shipped splits; training on one tree and testing on the other; a pooled random

TABLE II

SAME FEATURES, SAME CLASSIFIER, SAME LABELS; ONLY THE SPLIT PROTOCOL CHANGES. “GROUP LEAK” IS THE SHARE OF TEST IMAGES WHOSE DUPLICATE OR AUGMENTATION FAMILY ALSO APPEARS IN TRAINING. TEN SEEDS.

Protocol	Bal. acc.	Acc.	Group leak	n_{test}
<i>Six categories</i>				
Vendor split, imbalanced tree	0.325	0.661	26.7%	75
Vendor split, balanced tree	0.521	0.521	30.5%	226
Train one tree, test other	0.451	0.727	25.3%	75
Group-aware	0.665	0.705	0.0%	172
Pooled random	0.709	0.721	81.6%	339
<i>Three zones</i>				
Vendor split, imbalanced tree	0.497	0.751	26.1%	69
Vendor split, balanced tree	0.562	0.627	33.3%	189
Train one tree, test other	0.665	0.841	24.6%	69
Group-aware	0.727	0.765	0.0%	180
Pooled random	0.735	0.779	80.6%	293

split that ignores duplicates, which is what an ImageFolder plus train_test_split pipeline produces; and a group-aware split over connected components of the exact-duplicate, near-duplicate, and augmentation-family graph [22], [23].

Table II supports three claims, and refutes a fourth that we had expected to make.

The reported number is largely a choice. Across five defensible protocols on identical data, six-category balanced accuracy spans 0.325 to 0.709: a 38.5-point range attributable entirely to how the archive is read. A reader given only a single number cannot tell where in that band it sits.

Metric choice compounds it. Because the vendor’s splits are severely imbalanced, plain accuracy and balanced accuracy diverge sharply—0.661 versus 0.325 on the imbalanced tree’s own test split, a 33.6-point gap on one protocol with one model. Reporting accuracy on these splits is itself inflationary.

Zones are markedly more protocol-stable. The same five protocols span 23.8 points under the three-zone collapse against 38.5 under six categories, a 38% reduction in protocol-induced variance. The coarser target is not merely easier; it is less sensitive to a methodological choice the reader cannot see.

But duplicate leakage alone does not explain inflated results. The pooled random split leaks the duplicate or augmentation family of 81.6% of its test images and still exceeds the group-aware split by only 4.4 points (0.8 under zones). This is a negative result and we report it as one: in this archive, leakage is real but second-order. Critically, *no* protocol we could construct—including the most contaminated—approaches the 97.95–99.99% figures in the literature. Under this standard recipe the published numbers are not reachable by contamination alone, so they must originate elsewhere: in augmentation applied before splitting in downstream derivatives, in test sets of a few dozen images, or in selection among

TABLE III

CROSS-LABEL DUPLICATE GROUPS AT TWO LABEL GRANULARITIES. BOTH VIEWS USE THE SAME BYTE-IDENTICAL GROUPS.

	Six categories	Three zones
Conflicted duplicate groups	119	103
Conflicted files	327	280
<i>Conflicted files by class</i>		
Positive / neutral	5	1
Negative affect	292	279

many reported runs. Our audit constrains the explanation without completing it.

V. WHY THE LABELS CONTRADICT

The contradictions are not spread evenly across the label space. Mapping the six categories onto three valence/arousal zones—zone 1 (natural, joy), zone 2 (anger, fear), zone 3 (sadness)—and recounting gives Table III.

Two things follow. First, collapsing to zones absorbs 16 of the 119 categorical contradictions (13.4%): those are cases where annotators disagreed between anger and fear, or between labels that fall inside one zone, and the coarser schema simply does not record the disagreement. Second, and far more strikingly, the residue is almost entirely negative: 279 of 280 zone-level conflicted files are in zones 2 and 3, against one in zone 1.

Annotators agree, essentially perfectly, on whether an autistic child’s expression is positive or negative. They cannot agree on which negative emotion it is. That pattern is what [17] and [18] predict: if autistic expressions are produced in a register non-autistic raters find ambiguous, the ambiguity should concentrate where the categories are closest together, which is inside negative affect.

This reframes the contradictions. They are not simply sloppiness in one archive. A six-way forced choice over ambiguous expressions *manufactures* contradiction, because an annotator who perceives an ambiguous negative expression must nonetheless pick one box. The label schema is generating the noise the models are then scored against.

VI. BEYOND FACE-ONLY CATEGORICAL RECOGNITION

Two design consequences follow, and one caution.

Coarser, action-oriented labels. If the reliable signal is valence and the unreliable signal is the specific negative category, then a label space built on the reliable boundary is better founded than one built across it. This is the argument for Emotion Zones: classes defined by what a caregiver would *do*—de-escalate, engage, comfort—rather than by which Ekman category a non-autistic annotator would select. Zones are not merely a convenience; on these data they are the granularity at which annotation is reproducible.

TABLE IV

LEAKAGE-CONTROLLED REFERENCE RESULTS.

GROUP-AWARE SPLIT, 251 TEST IMAGES IN 181 DUPLICATE GROUPS, 20 SEEDS, CLUSTER BOOTSTRAP OVER GROUPS.

Model	Bal. acc.	95% CI	κ	Zone-3
Majority class	0.333	—	0.000	0.000
DeepFace (NT-trained)	0.377	[0.331, 0.425]	0.115	0.058
Linear probe	0.731	[0.669, 0.794]	0.638	0.606
MLP probe	0.778	[0.725, 0.834]	0.710	0.653

The caution: do not credit the chance floor. A collapse from six classes to three raises the chance baseline from 1/6 to 1/3, and comparing raw balanced accuracies across the two spaces silently rewards that. On our earlier data a commercial classifier moved from 28.17% (six categories) to 45.12% (three zones), an apparent 16.95-point gain. Chance-corrected, the same relabeling is worth 3.88 points. The zone argument must rest on construct validity and annotation reliability, which the evidence above supports, not on a discrimination gain that is mostly an artifact of counting. We therefore report Cohen’s κ alongside balanced accuracy throughout.

Context beyond the face. Table IV gives our leakage-controlled reference results on a group-aware split with 20 training seeds and a cluster bootstrap over duplicate groups. Two numbers matter. A neurotypically trained commercial classifier reaches 37.7% balanced accuracy with a 95% interval of [33.1, 42.5] against a majority-class floor of 33.3%, and its recall on zone 3 is 5.8%: on this test set it is close to indistinguishable from predicting the majority class. A probe trained on the same population reaches 77.8%. The gap is not that the task is impossible; it is that a face-only model trained on non-autistic faces is the wrong instrument.

This is where multimodal and vision-language approaches earn their place, and the reason is a data argument rather than an accuracy argument. Training a face-only autism-specific model requires autism-specific face data, which is precisely the resource whose acquisition produced the retraction wave. A vision-language model brings broad prior knowledge about scenes, activities, and posture without requiring any autism-specific training corpus, and can supply the environmental and interaction context that a cropped face cannot. Given that our clean face-only ceiling is 77.8% and that [16] argues facial configuration underdetermines emotional state in principle, the remaining headroom is unlikely to be found in better face classifiers. It is more likely to be found in context—and in reporting a zone with an explanation a caregiver can check, rather than a category with a confidence score.

VII. LIMITATIONS AND ETHICS

This is an audit of one archive and does not establish prevalence across the field, though [15] reports consistent qualitative findings. Our duplicate detection is exact-hash and perceptual-hash based; it will miss re-encoded or cropped

duplicates, so our rates are lower bounds. Cluster bootstrap intervals are percentile intervals, which can under-cover [24]. We use no subject identities because the archive supplies none, so our groups are duplicate and augmentation families, not people; a subject-disjoint split would likely be stricter still.

We report only aggregate statistics. No image, filename, hash, or per-image prediction is released. We make no claim that any depicted child is autistic, that any label reflects a felt emotion, or that any system built on this analysis should be deployed. The archive’s provenance is disputed and its license is unstated; we redistribute nothing and cite the source so the audit can be reproduced from it. Under the EU AI Act, emotion inference in education is prohibited except under a medical exemption whose justification explicitly references autism, which raises rather than lowers the evidentiary bar on the benchmarks used to support such systems.

VIII. CONCLUSION

The most widely used public autism emotion archive ships two overlapping copies sharing 45% of the smaller, is 21.4% duplicated, and contains 119 groups of pixel-identical images with contradictory labels. Varying only the split protocol moves six-category balanced accuracy across a 38.5-point band, so a single reported number on this archive carries little information about a model. Duplicate leakage is not the main culprit—it is worth 4.4 points—and no protocol we could construct reaches the 97.95–99.99% figures in the literature, which we therefore cannot attribute to contamination alone.

The label contradictions concentrate almost entirely in negative affect, which indicates that the six-category schema is manufacturing them rather than merely recording them, and the three-zone collapse cuts protocol-induced variance by 38%. Coarser action-oriented zones and multimodal context are better founded responses than finer face-only categories. Any future benchmark in this area should report a duplicate audit, a group-aware split, and chance-corrected metrics as a condition of being believed—and, given how this archive was assembled, should not be built by scraping at all.

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